

COURSE TITLE : CAD Lab I
COURSE CODE : 4188
COURSE CATEGORY : A
PERIODS/WEEK : 3
PERIODS/SEMESTER : 39
CREDITS : 2

TIME SCHEDULE

Module	Topics	Period
1	Introduction to computer aided drafting	6
2	CAD Basics	6
3	Working with CAD Software	18
4	Developing 2D drawing	9
TOTAL		39

Course Objective :

Sl.	Sub	Student will be able to
1	1	Identify CAD system components
	2	Use CAD for 2D drawings
	3	Prepare simple plans, sections and elevations in CAD
Total periods per semester		45

After completing this course student should be able to:

UNIT – I: Introduction to computer aided drafting

1.1.0. Understand Auto CAD

- 1.1.1. Know CAD system components
- 1.1.2. Know CADD work station
- 1.1.3. Identify elements of drawing window
- 1.1.4. Know application of CAD
- 1.1.5. Know development of CAD
- 1.1.6. Identify system requirements

UNIT – II: CAD Basics

2.1.0. Understand CAD basics

- 2.1.1 Know start up dialogue box options
- 2.1.2. Identify pointing devices like mouse
- 2.1.3. Describe accessing command
- 2.1.4. Use a template
- 2.1.5. Open a drawing sheet
- 2.1.6 Enter data.
- 2.1.7. Select entity and correct mistakes
- 2.1.8. Set limits
- 2.1.9. Draw lines, circles, Arcs etc.

UNIT – III: Working with CAD Software

3.1.0. Work using CAD

- 3.1. 1.Edit using commands like erase, copy, trim, extend, fillet etc
- 3.1.2. Dimension a drawing
- 3.1.3. Add text in a drawing

UNIT – IV: Developing 2D drawing

4.1.0. Familiarise with draw commands and edit commands

- 4.1.1. Know more draw commands like point, trace, sketch, solid, hatch, poly-line etc
- 4.1.2. Know object snap settings
- 4.1.3. Use more edit commands like chamfer, break, stretch, scale, array etc
- 4.1.4. Use layer commands
- 4.1.5 Make and insert blocks
- 4.1.6. Retrieve the drawing as hard copy & soft copy
- 4.1.7 Draw layouts and plot
- 4.1.8 Draw detailed plan of small buildings

COURSE CONTENTS

MODULE – I –INTRODUCTION TO CAD

Drawing using CAD – advantages of using CAD-CAD system components- CADD work station--elements of drawing window--application of CAD-- development of CAD--system requirements
Drawing by CAD Software – Drawing units zooming and panning – plotting

UNIT – II: CAD Basics

Start up dialogue box – options-- CAD Software initial screen, menu bar, tool bars, graphic window, command line, status bar-- Pointing devices like mouse-- Accessing commands – command line pull down menu – using commands transparently, getting help from CAD Software-- Use a template-open a drawing-- Data entry, entity selection, correcting mistakes-- Setting limits-- Drawing lines, circles, Arcs etc- Simple exercises

UNIT – III: Working with CAD Software

Editing commands like erase, copy, trim, extend, fillet etc--Adding dimensions and text-- Drawing simple geometrical figures with CAD Software—

UNIT – IV: Developing 2D drawing

Draw commands like point, trace, sketch, solid, hatch, poly-line etc- Object snap settings-- More edit commands like chamfer, break, stretch, scale, array etc-- Layer manipulation-- Making and inserting blocks-- Retrieving the drawing as hard copy & soft copy-- Drawing layouts and plotting-- Drawing detailed plan of small buildings

REFERENCE BOOKS

1. Dayanithi - AutoCAD
2. Sham Tickoo - AutoCAD with Application -